$S, \bar{S}$

$\varphi: S \to \bar{S}$ 对于 $p \in S$, $\omega_1, \omega_2 \in T_pS \longrightarrow \langle \omega_1, \omega_2 \rangle_p = \langle d\varphi_p(\omega) \cdots \rangle_{\varphi(p)}$

$x: U \to S$ E, G, F.

$\bar{x}: U \to \bar{S}$ $\bar{E}, \bar{G}, \bar{F}$.

① $x(u,v) = (a\cosh v \cos u, a\cosh v \sin u, av)$ $E = a^2\cosh^2 v$, $F = 0$, $G = a^2\cosh^2 v$

② $\bar{x}(\bar{u}, \bar{v}) = (\bar{v}\cos\bar{u}, \bar{v}\sin\bar{u}, a\bar{u})$ $0 < \bar{u} < 2\pi$, $-\infty < \bar{v} < +\infty$

$(\bar{u} \in \mathbb{R})$

变换 $\Longrightarrow \bar{u} = u$ $u \in (0, 2\pi)$

$\bar{v} = a\sinh v$ $v \in \mathbb{R}$

$\bar{x}(u,v) = (a\sinh v \cos u, a\sinh v \sin u, au)$ $\bar{E}, \bar{F}, \bar{G}$.

Def. $\langle d\varphi_p(v_1), d\varphi_p(v_2) \rangle = \lambda^2(p) \langle v_1, v_2 \rangle$.

α, β $\cos\theta = \frac{\langle \alpha, \beta \rangle}{\|\alpha\| \cdot \|\beta\|}$ 不变

S 可定向 X_u, X_v, N

	X_u	X_v	N
X_{uu}	Γ_{11}^1	Γ_{11}^2	L_1
X_{uv}	Γ_{12}^1	Γ_{12}^2	L_2
X_{vv}	Γ_{22}^1	Γ_{22}^2	L_3
N_u	a_{11}	a_{21}	
N_v	a_{12}	a_{22}	

$L_i \longrightarrow \underline{\langle X_{uu}, N \rangle} = L_1$ L_2 L_3

e

$E_u = (X_u, X_u)_u = 2\langle X_{uu}, X_u \rangle = 2\Gamma_{11}^1 E + 2\Gamma_{11}^2 F$ $EG - F^2 \neq 0$.

$\Gamma_{11}^1 F + F_{11}^2 G$

$\langle X_{uu}, X_v \rangle = F_u - \frac{1}{2} E_v$

$(X_{uu})_v = (X_{uv})_u$ $\quad A_1 X_u + B_1 X_v + C_1 N = 0$

$(X_{vv})_u = (X_{vu})_v \Longrightarrow$ {

$N_{uv} = N_{vu}$

$A_1 = 0 \Rightarrow \Gamma_{11}^1 \Gamma_{12}^2 + \Gamma_{11}^2 \Gamma_{22}^2 + e a_{22} + (\Gamma_{11}^2)_v = \Gamma_{12}^1 \Gamma_{11}^2 + \Gamma_{12}^2 \Gamma_{12}^2 + f a_{12} + (\Gamma_{12}^2)_u$

$\Gamma_{ij}^k l_{ij} = f a_{21} - e a_{22}$

$= f \frac{eF - fG}{EG - F^2} - e \frac{fF - gG}{EG - F^2}$

$= \frac{-f^2 G + gGe}{EG - F^2} = -EK$

$\omega \quad p \in U. \quad Y \in T_p S$

$\alpha : (-\varepsilon, \varepsilon) \to U. \quad \alpha(0) = p \quad \alpha'(0) = Y$